

33S1 — Dynamic Interface Promotion and Recursive Boundary Closure

Professional reading edition of the evolving public MKUFT canon

Mark Charles McLaughlin

ORCID: [0009-0005-7736-1511](https://orcid.org/0009-0005-7736-1511)

Scientific affiliation	Independent Researcher
Object status	Research module reading edition — not a standalone Zenodo publication
Public source	Open canonical public source
Source commit	26e06b6fea52e6b052dbdd540e80d1d926b96b07
MKUFT backbone DOI	10.5281/zenodo.17780566

Status boundary. This PDF is a reader-facing rendering of an evolving GitHub research module. It is not a frozen standalone paper and does not receive a separate DOI. The cited Git commit fixes the exact source identity of this edition.

Rights. Copyright © 2026 Mark Charles McLaughlin. All rights reserved. Historical licences remain controlling only for their exact frozen objects.

1. Purpose

Module 33 already allows boundary/context data, relation scopes and history to contribute to the typed admissibility descriptor

$$\Xi_t = \mathcal{Q}_t(\mathcal{U}_t, E_t),$$

and already readdresses the next state after physical propagation and registration. Module 22A separately requires every load-bearing relation exposed by analysis to become a typed object, while Module 32S4 allows nested contextual constraints and changing boundary conditions to alter physical realisation.

One ambiguity remains operationally important:

When should an interface remain bundled inside ordinary context, and when has it earned explicit state-variable status inside the recursive update?

This module supplies that promotion criterion. The result is a refinement of the closed Module 33 drivetrain, not a replacement for it.

2. Typed interface object

Let A_t and B_t be separately addressed systems or subobjects participating in one declared joint model. Let

$$\boxed{\mathcal{J}_{AB,t}}$$

denote a typed **interface/junction state** between them.

The symbol $\mathcal{J}$ is used deliberately so that it is not confused with the admissible path family Γ^{adm} already used in Module 33.

A candidate interface may be generated by

$$\boxed{\mathcal{J}_{AB,t} = J_{AB}(A_t, B_t, E_t, H_t),}$$

where the implementation must declare what A , B , E , H , and $\mathcal{J}$ physically or operationally mean.

$\mathcal{J}_{AB,t}$ is not assumed to be a spatial surface. Depending on the domain it may encode a physical boundary condition, coupling state, junction relation, shared control condition, membrane state, contact geometry, or another explicitly typed interface. Cross-domain similarity does not establish one common mechanism.

3. Dynamic-interface promotion gate

An interface is not promoted merely because it is visually striking, mathematically elegant, or easy to name.

Promotion requires all three conditions below.

3.1 Operational definition and trackability

The candidate interface state must be operationally defined and measurable, manipulable, inferable, or otherwise trackable to the resolution needed by the claim. It must be possible to say what counts as the same interface state, a changed interface state, and the uncertainty of that comparison.

This does **not** require $\mathcal{J}$ to be causally, ontologically, or microscopically independent of A , B , environment or history. A fully P-recoverable interface can still be a useful explicit macrostate or sufficient state variable. Independence is a separate $I \rightarrow P$ question handled by Module 33.

If no stable typed interface quantity can be operationally recovered at the resolution of the claim, retain the relation inside the simpler bundled context.

3.2 Load-bearing prospective effect

Under matched controls for the relevant A , B , E and H variables, a controlled change

$$\mathcal{J}_{AB,t}^{(1)} \longrightarrow \mathcal{J}_{AB,t}^{(2)}$$

must prospectively change a declared downstream object such as the admissibility descriptor

$$\Xi_t,$$

the active P-law object

$$\mathcal{L}_{P,t} = (\mathcal{D}_{P,t}, \mathcal{T}_{P,t}, \mathcal{W}_{P,t}),$$

or a preregistered higher-scale realisation/readout.

A change in notation or in an intermediate representation is insufficient. The deformation must alter a load-bearing prediction, admissible support, transition rule, weighting, capability, or outcome under the declared model.

3.3 Recursive back-action

The realised transition must in turn update the interface in a way that can matter on the next cycle. Write schematically

$$\mathcal{J}_{AB,t+\Delta} = R_{\mathcal{J}}(\mathcal{J}_{AB,t}, A_{t+\Delta}, B_{t+\Delta}, O_{t+\Delta}, E_{t+\Delta}, H_{t+\Delta}).$$

The updated interface then re-enters the next admissibility calculation.

The minimal recursive pattern is therefore

$$\mathcal{J}_{AB,t} \rightarrow (A_{t+\Delta}, B_{t+\Delta}) \rightarrow \mathcal{J}_{AB,t+\Delta} \rightarrow \Xi_{t+\Delta}.$$

If the final arrow carries no prospective consequence, the recursion has not earned explicit state-variable status.

4. Promotion does not create a fifth SIPO layer

When the gate passes, the correct move is **typing refinement**, not ontology inflation.

The interface can be represented explicitly as a named subobject of the appropriate existing address/relation structure, for example inside I_t or b_t , or an implementation may expose it as an explicit coordinate of its working addressed state:

$$\tilde{\mathcal{U}}_t = (\mathcal{U}_t, \mathcal{J}_{AB,t}).$$

This notation means that the implementation has made a previously bundled load-bearing relation explicit. It does not define a universal fifth fundamental layer.

Compressed rule:

If a controlled interface state changes the next admissible law object and the realised transition changes that interface, the interface must re-enter the next addressed state.

5. Joint update and recursive boundary closure

For a declared two-system model, write the joint addressed object schematically as

$$\mathcal{U}_t^{AB} = (\mathcal{U}_t^A, \mathcal{U}_t^B, \mathcal{J}_{AB,t}, H_t^{AB}).$$

The capstone architecture then applies without changing its parent structure:

$$\Xi_t^{AB} = \mathcal{Q}_t^{AB}(\mathcal{U}_t^{AB}, E_t),$$

$$\mathcal{L}_{P,t}^{AB} = \mathfrak{C}_P^{AB}(\Xi_t^{AB}, P_t^{AB}, E_t),$$

followed by physical propagation, a correctly typed $P \rightarrow (P, O)$ instrument where measurement occurs, and readdressing of the joint state including $\mathcal{J}_{AB,t+\Delta}$.

Thus a load-bearing interface is not scenery around the update. It is one of the typed conditions from which the next update is constructed.

6. Scale-role transition of a boundary

A boundary/interface can change **role** with scale without changing its lower-scale identity by fiat.

Suppose A_t , B_t , and $\mathcal{J}_{AB,t}$ jointly pass the Module 32S promotion tests for a higher-scale effective unit. Then define a higher-scale readout

$$C_t = R_{AB \rightarrow C}(A_t, \mathcal{J}_{AB,t}, B_t).$$

At the lower scale, $\mathcal{J}_{AB,t}$ is a boundary/junction between distinguishable constituents. At the promoted scale, that same typed relation may become **internal structure of C_t** .

The resulting role sequence is

boundary at scale $L \rightarrow$ load-bearing stateful relation at $L \rightarrow$ internal relation of a promoted whole at $L + 1$.

This is not automatic. If the joint object fails the scale-promotion tests, no higher-scale object is inferred merely because a boundary exists.

7. Geometric bridge comparator — strictly bounded

Einstein and Rosen's 1935 construction is useful here only as a **geometric comparator**. Their original paper represents physical space by two identical sheets connected by a bridge. DOI [10.1103/PhysRev.48.73](https://doi.org/10.1103/PhysRev.48.73).

The comparator illustrates one narrow point:

two regions that are distinct under one description can participate in one larger geometry whose connecting structure is itself part of the complete object.

No stronger inference is inherited.

In particular, $\mathcal{J}_{AB,t}$ is **not** thereby identified with an Einstein–Rosen bridge, a spacetime wormhole, a throat, a traversable shortcut, or an independent I-layer structure. MKUFT does not obtain evidence for literal wormholes from this comparison.

The bridge picture motivated an interface question; the promotion gate above is the canonical result.

8. Physical boundary dynamics and energy accounting

Module 32S4 already records established P-layer cases in which boundary conditions alter allowed modes and dynamical boundary modulation can produce measurable excitations. Those comparators support the general legitimacy of stateful boundary variables inside physical models; they do not establish an independent I layer.

Where $\mathcal{J}_{AB,t}$ is a physical boundary or control parameter and is actively changed, all work, drive, preparation, thermal, chemical, electromagnetic, mechanical and measurement inputs associated with that change remain part of the physical ledger.

Therefore:

promoting a boundary into the state makes its energetic bookkeeping more explicit; it never licenses omission of the energy required to create, move, modulate or maintain that boundary.

No claim of self-power, free vacuum energy or zero-point extraction follows from boundary-conditioned mode changes. Any such claim would require a closed-cycle experiment showing a reproducible residual after complete input accounting and uncertainty analysis.

9. P-only and I→P classification

Passing the dynamic-interface gate establishes only that the interface is load-bearing for the declared model.

If an adequate physical account of A , B , $\mathcal{J}$, environment and history predicts the observed change, the result remains P-realised and belongs at Module 33 Level 2 or below.

Only if a separately specified I-variable adds held-out or interventional prediction beyond that strongest adequate P-account does the independent I→P question advance. The interface itself must not be relabelled as I merely because it is relational.

10. Failure and reduction rules

Do not promote the interface when:

1. no stable typed $\mathcal{J}$ can be operationally defined or tracked at the resolution of the claim;
2. controlled deformation of $\mathcal{J}$ changes no preregistered downstream admissibility, law object, capability or outcome;
3. the apparent effect disappears when ordinary boundary, hysteresis, drift, memory, geometry or field-history variables are controlled;
4. the interface is updated by the system but its updated value adds no prospective prediction to the next cycle;
5. a simpler P-only model captures the same behaviour with equal or better held-out performance;
6. the proposed higher-scale whole fails the existing Module 32S promotion tests;
7. a physical boundary-energy claim omits drive, preparation, maintenance or measurement inputs;
8. a geometric analogy is used as evidence for a mechanism in another layer.

Reduction rule:

If explicit interface promotion adds no conclusion-changing discrimination beyond the already adequate addressed state, retain the simpler representation.

11. Contribution boundary

This module does not claim invention of boundaries, junction conditions, interface dynamics, coupled-system state variables, recursive systems, Einstein–Rosen bridges, scale coarse-graining, or dynamical boundary physics.

The candidate MKUFT contribution is the operational conjunction:

combine recursive-address closure with the Module 33 law-object update so that a relation/interface is promoted from bundled context to explicit dynamic state only when it is operationally trackable, prospectively load-bearing for admissibility or the active law object, recursively changed by realised dynamics, and capable—after independent scale-promotion tests—of changing role from lower-scale boundary to higher-scale internal structure.

Historical priority for that exact conjunction is not asserted without broader review.

12. Compressed canonical rule

Precise form:

A boundary/interface remains context until controlled deformation shows that its state changes a load-bearing downstream object. If the realised transition then changes that interface and the changed interface alters the next admissibility calculation, promote it into the recursively addressed state. If the coupled whole later passes scale-promotion tests, the former boundary may become an internal relation of the higher-scale object.

Mnemonic:

If the interface carries the future, the interface belongs in the state.